Chapter 3.
The Language of Mathematics

Elementary Set Theory

Set

- The concept of set is basic to all of mathematics and mathematical applications.
- All of mathematics, as well as subjects that rely on mathematics, including computer science and engineering, make use of some of the fundamental concepts of mathematics such as sets.
- Discrete mathematics is concerned with structures such as graphs (sets of vertices and edges) and Boolean algebras (sets with certain operations defined on them).

Set

- a collection of objects.
- each object in a set is called an **element** or **member** of the set. The elements or objects of the set are enclosed by a pair of braces { }.
- notations:
  - capital or uppercase letters are usually used to denote sets while small or lowercase letters denote elements of a set.
  - $\in$ denotes "is an element of" or "belongs to“
  - $\notin$ denotes "is not an element of" or "does not belong to”

Examples of Set

- let A - the set of letters in the English alphabet

  B - the set of primary colors

  C - the set of positive integers

  – $g \in A$ $\rightarrow$ g is a member of set A
  – white $\notin$ B $\rightarrow$ white is not a member of set B
  – $100 \in C$ $\rightarrow$ 100 is a member of set C

Set Builder

- There are two ways to describe a set.
  - the **list (or roster) method** describes a set by enumerating the elements of the set
  - the **rule method (or set builder)** describes the set by a statement or a rule.

Set Builder

- Examples:
  - Using the rule method the given sets can be defined as:
    - $A = \{x \mid x \text{ is an English alphabet}\}$
    - $B = \{a \mid a \text{ is a primary color}\}$
    - $C = \{y \mid y \text{ is a positive integer}\}$
  - Using the roster method, we have:
    - $A = \{a, b, c, d, e, \ldots, z\}$
    - $B = \{\text{red}, \text{blue}, \text{yellow}\}$
    - $C = \{1, 2, 3, 4, 5, \ldots\}$

Cardinality of a Set

- **Definition.** The **cardinality** of set A, or the cardinal number of A, denoted as **n(A)**, is the number of elements in A.
- **Example.**

Using the given sets

$$n(A) = 26,\ n(B) = 3 \text{ and } n(C) = \infty.$$

Cardinality of a Set

- **Definition**. A set is **finite** if there is one counting number that indicates the total number of elements in the set. It is **infinite** if in counting the elements, we never come to an end.

- **Example.** Using the given sets,
  - A is a finite set because it has 26 elements;
  - B is also finite
  - C is infinite

Null and Singleton Sets

- **Definition.**

- The **null set** or **empty set**, denoted by the symbol ϕ, is the set that contains no elements, that is, A is empty if and only if (iff) $n(A) = 0$.

- A **singleton set** is a set that contains only one element, that is, B is a singleton set if and only if $n(B) = 1$.

Null and Singleton Sets

- Example.
  - let $D = \{ x \mid x$ is a month in the Gregorian Calendar with less than 28 days$\}$
  - let $E = \{y \mid y$ is a prime number, $24 < y < 30\}$

$\rightarrow$ $n(D) = 0$, so $D = \phi$.

$\rightarrow$ $E = \{29\}$. $n(E) = 1$,

so E is a singleton set.

Subsets

- **Definition.** Set A is a subset of B, denoted as $A \subseteq B$, iff every element of A is also an element of B.

- **Law 3.1**
  - Every set is a subset of itself, that is, $A \subseteq A$, for any set A.
  - The null set is a subset of any set, that is, $\phi \subseteq A$, for any set A.

Subsets

- **Definition.** Set B is a **proper subset** of A, denoted as $B \subset A$, if B is a subset of A and there is at least one element of A that is not in B. That is, $B \subset A$ iff $B \subseteq A$ and $B \neq A$.

Examples of Subsets

Let G = {x/x is an integer}.

Let F = {x/x is a whole number}

- Then, C $\subseteq$ G, where C as defined earlier is the set of positive integers, because every element of C is found in G, more exactly, C $\subset$ G
- However, F $\not\subset$ C because 0 $\in$ F but 0 $\notin$ C.
- But, C $\subset$ F.

Universal Set

- **Definition.** The set containing all of the elements for any particular discussion is called the **universal set** .
- Remark: The universal set U does not contain everything, only all things pertinent to the discussion.

Universal Set

- Example.

  Let

  H = { x | x is an even number greater than 5 but less than or equal to 20}

  I = { y| y is a letter in the word “Mississippi”}.

  – Then, we can define the following universal sets corresponding to H and I respectively.

  U_1 = {x/x is a natural number}

  $\rightarrow$ $H \subset U_1$.

  U_2 = {y/y is a letter in the English alphabet}

  $\rightarrow$ $I \subset U_2$.

Operations on Sets

- Intersection of sets
- Union of sets
- Difference of sets
- Complement of a set
- Cartesian product of two sets

Set Union and Intersection

- **Definition.** The **intersection** of two sets A and B, denoted as $A \cap B$, is the set consisting of all elements common to both or belonging to both A and B. In symbol,

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Or, we can write $x \in A \cap B \leftrightarrow (x \in A \wedge x \in B)$.

Set Union and Intersection

- **Definition.** The **union** of sets A and B denoted as $A \cup B$ is the set of all elements which belong to A or B. In symbol,

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Or, we can write $x \in A \cup B \leftrightarrow (x \in A \vee x \in B)$.

Set Difference and Complement

- **Definition.** The **difference** of sets A and B is the set of elements which belong to A but do not belong to B. We denote the difference of A and B as A - B, which is read as "A difference B" or "A minus B". Thus,

$$A-B = \{x \mid x \in A \text{ and } x \notin B\}.$$

Or, we can write $x \in A - B \leftrightarrow (x \in A \wedge x \notin B)$.

Set Difference and Complement

- **Definition.** The **complement** of set A, denoted as A' (read "A prime") is the set of elements found in the universal set that are not in A. In symbol,

$$A' = \{x/x \in U \text{ and } x \notin A\}.$$

Or, we can write $x \in A' \leftrightarrow (x \in U \wedge x \notin A)$.

Cartesian Product

- **Definition.** The **Cartesian product** of two sets A and B denoted by A x B (read as "A cross B") is the set of ordered pairs (x, y) that can be formed such that x is an element of A and y is an element of B. In symbol

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}.$$

Or, we can write $(x, y) \in A \times B \leftrightarrow (x \in A \wedge y \in B)$.

Cartesian Product

- **Remark.** An **ordered pair** is a pair of elements (x, y) with x specified as the first element and y the second element.

Example

$U = \{ 1, 2, 3, 4, 5 \}$

$A = \{ 1, 2, 3 \}$

$B = \{ 2, 4 \}$

a.) $A \cap B = \{ 2 \}$

b.) $A \cup B = \{ 1, 2, 3, 4 \}$

c.) $A - B = \{ 1, 3 \}$

d.) $A' = \{ 4, 5 \}$

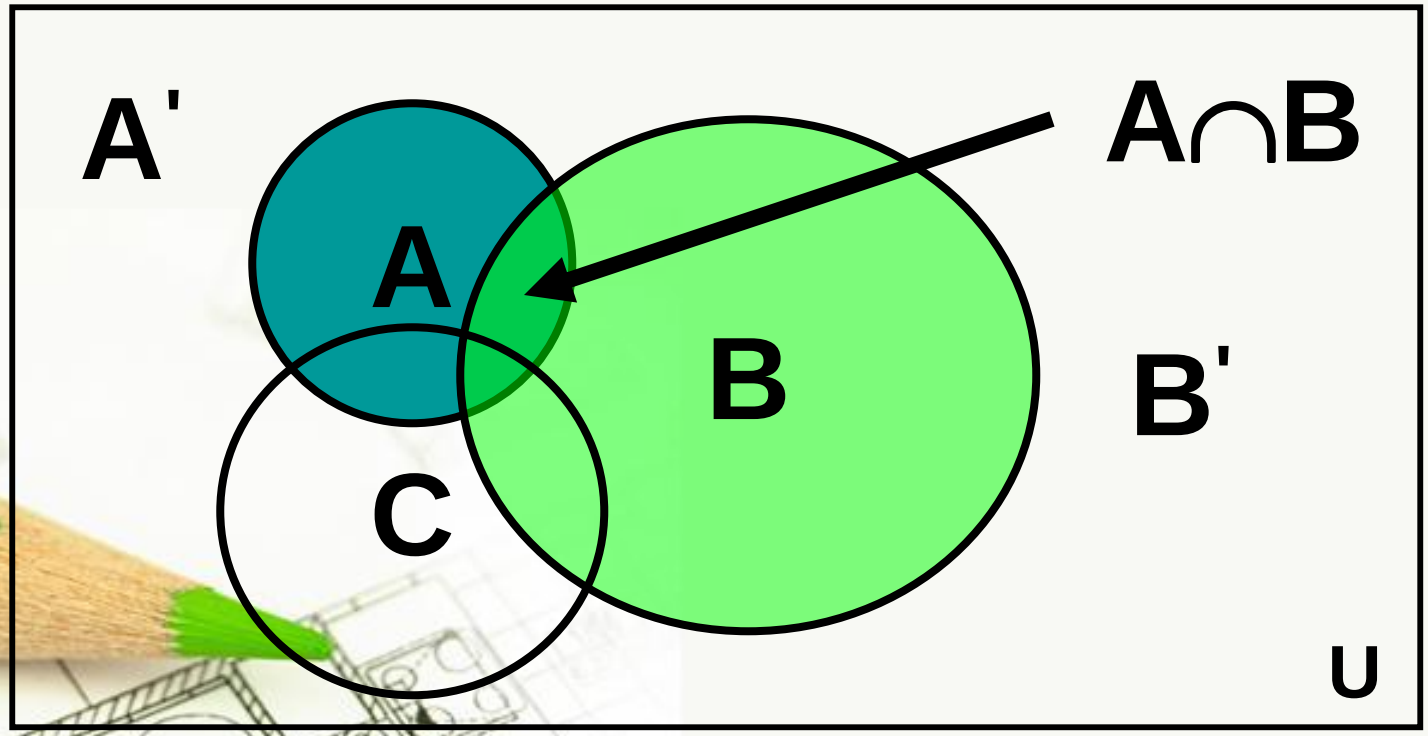
e.) $A \times B = \{ (1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4) \}$

Venn Diagram

- Represents the universal set U through a rectangle.
- Any set A belonging to U (or having U as the universal is represented by a closed region, usually circular, within the rectangle.
- The items (or points) inside and on the circle represent elements of A, while those outside represent objects not in A (complement of A).

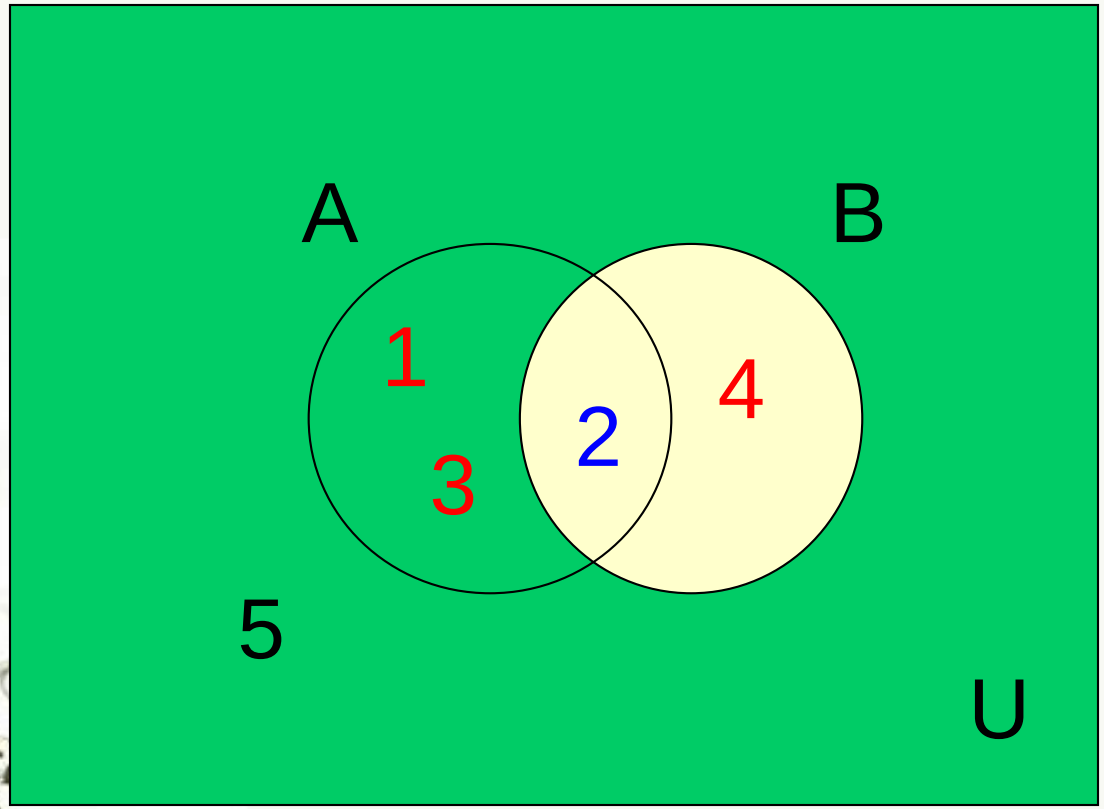
Venn Diagram

- used to depict the various unions, subsets, complements, intersections of sets.



Venn Diagram

U = { 1, 2, 3, 4, 5 }
A = { 1, 2, 3 }
B = { 2, 4 }



Equality of Sets

- Definition: Two sets X and Y are equal and we write X = Y if X is a subset of Y and Y is a subset of X, that is, X = Y iff $X \subseteq Y$ and $Y \subseteq X$. To put it another way, X = Y if they have exactly the same elements, that is, whenever $x \in Y$, then $x \in X$.
- Example :

  If $A = \{ x \mid x^2 + x - 6 = 0 \}$, $B = \{ 2, -3 \}$, then A = B

Law 3.2 (Properties of Set Union)

For any sets A, B, and C.

1. The union of any set with the null set is the set itself. That is, $A \cup \phi = A$.
2. The union of any set with itself is the set itself. That is, $A \cup A = A$
3. Set Union is commutative, that is, $A \cup B = B \cup A$.

Law 3.2 (Properties of Set Union)

4. Set Union is associative, that is, $(A \cup B) \cup C = A \cup (B \cup C)$.
5. Any set is a subset of its union with another set, that is, $A \subseteq A \cup B$.

Law 3.3 (Properties of Set Intersection)

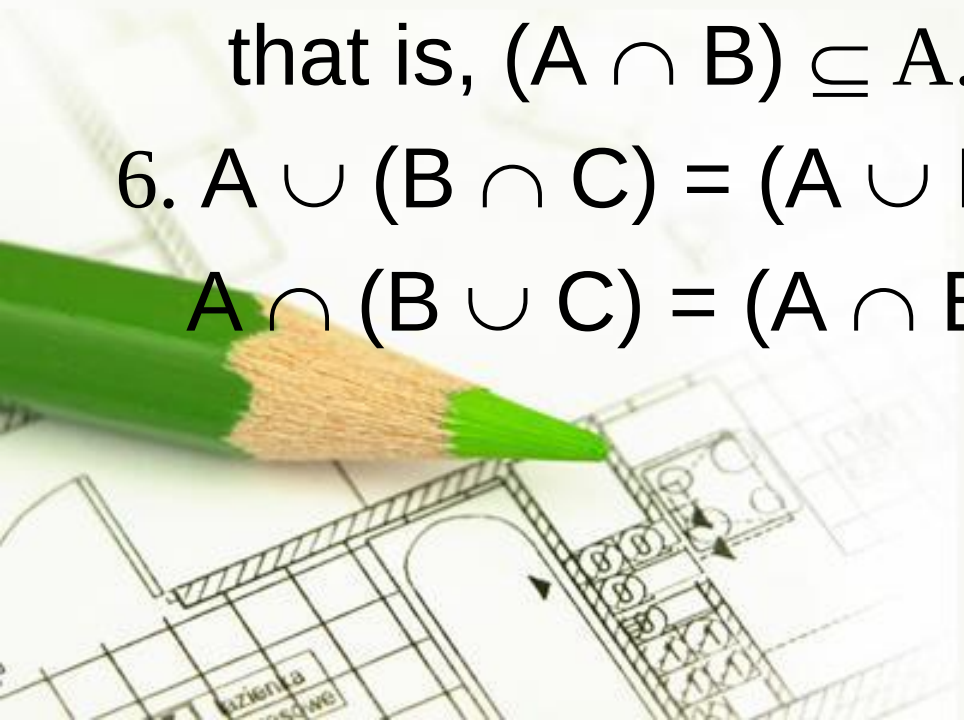
For any sets A, B, and C,

1. The intersection of any set with the null set is the null set. That is, $A \cap \phi = \phi$
2. The intersection of any set with itself is the set itself. That is, $A \cap A = A$.
3. Set intersection is commutative, that is, $A \cap B = B \cap A$.

Law 3.3 (Properties of Set Intersection)

4. Set intersection is associative, that is, $(A \cap B) \cap C = A \cap (B \cap C)$
5. The intersection of any given set with another set is a subset of the given set, that is, $(A \cap B) \subseteq A$.
6. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

   $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$



Law 3.4 (Properties of Set Difference)

For any set A, B, C,

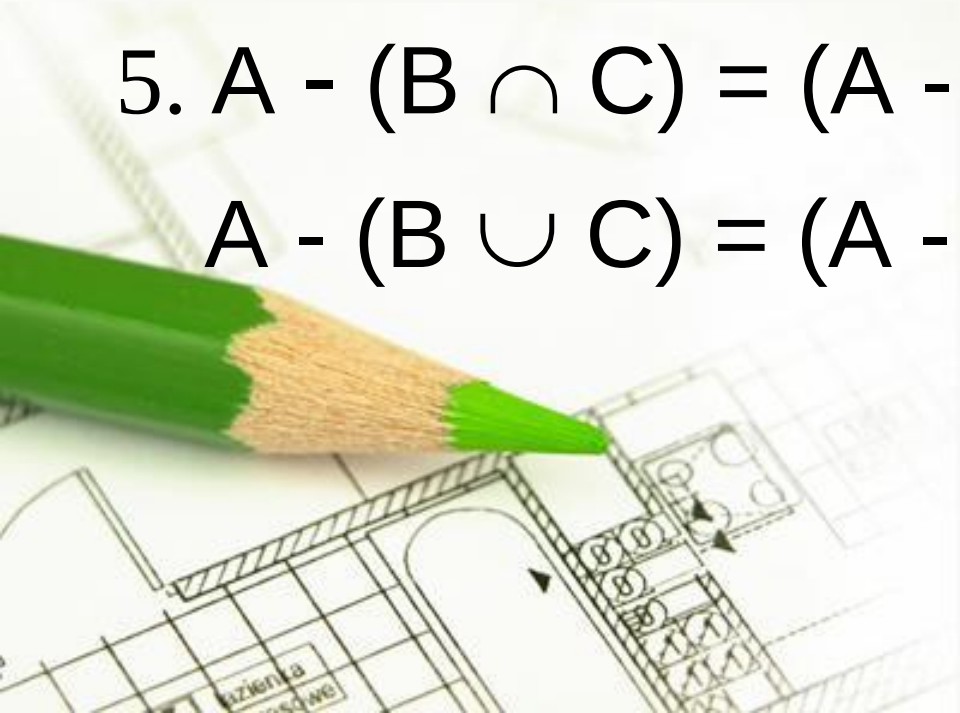
1. The removal of the null set from any set has no effect on the set. That is, $A - \phi = A$.
2. The removal of the elements of any set from itself will leave the empty set. That is, $A – A = \phi$.
3. No elements can be removed from the null set. That is, $\phi - A = \phi$.

Law 3.4 (Properties of Set Difference)

4. The result of removing the elements of a set from any given set is a subset of the given set, that is, $A - B \subseteq A$.

5. $A - (B \cap C) = (A - B) \cup (A - C)$

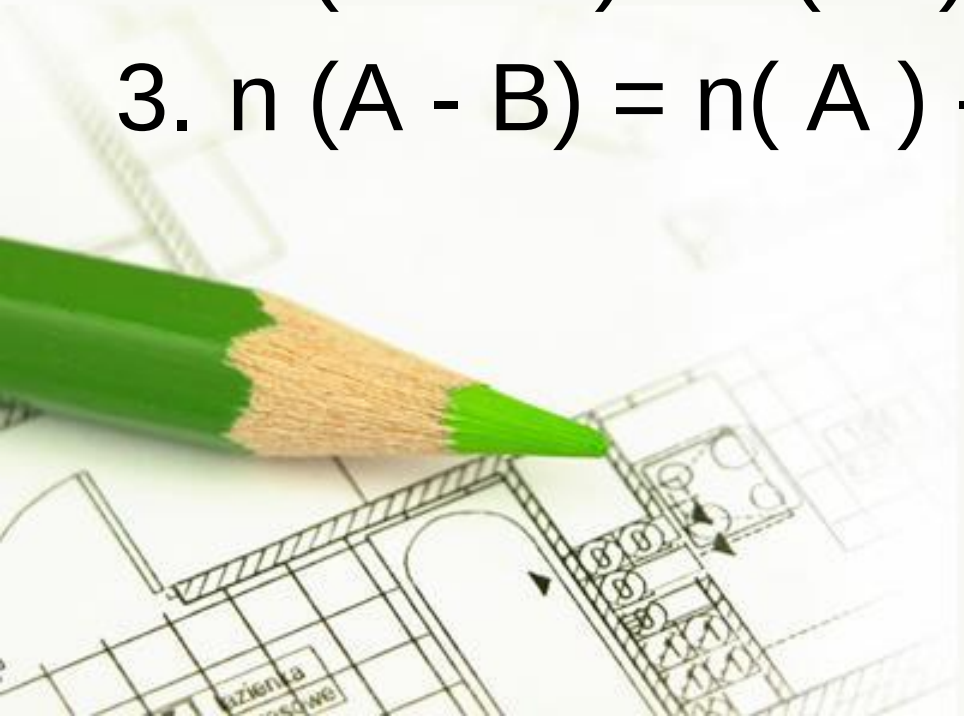
   $A - (B \cup C) = (A - B) \cap (A - C)$



Law 3.5

For any sets A, B, and C,

1. $n(A \cap B) = n(A) - n(A - B)$
2. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
3. $n(A - B) = n(A) - n(A \cap B)$



Power Set

- Definition. The set of all subsets (proper or not) of a set X, denoted $\wp(X)$, is called the **power set** of X.
- A method that we can consider in listing all subsets of a given set is to start with subsets of cardinality 0. In all cases, there is only one, the null set. Then, we consider all subsets of cardinality 1.

Power Set

- Next, we consider all subsets of cardinality 2. This process continues until subsets of cardinality equal to that of the given set is considered. Again, there will only be one subset of this cardinality: the original set itself.

Power Set

- **Example 1**

  If $A = \{a, b, c\}$

  $$\wp(A) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

  All but $\{a, b, c\}$ are proper subsets of A.

  For this example,

  $$n(A) = 3, \qquad n(\wp(A)) = 2^3 = 8.$$

Power Set

- **Law 3.6 (Properties of Power Sets)**

  For any set A, $A \in \wp(A)$.

  For any set A, $n(\wp(A)) = 2^{n(A)}$.